KOLMOGOROV, QUASI-SOBER AND SOBER PROPERTIES PASS TO LOCALLY CLOSED SUBSPACES

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Allowed standard prerequisites (editorial).

- The Kolmogorov, quasi-sober and sober separation properties, and the notion of a locally closed subspace. Set up in the author's sections on irreducible components and sober spaces, immediately before the counted statement. Fixes the three properties the statement transfers to a subspace and the hypothesis under which the transfer is claimed.

Lemma 0.1. *Let X be a topological space and let $Y \subset X$.*

(1) *If X is Kolmogorov then so is Y.*
(2) *Suppose Y is locally closed in X. If X is quasi-sober then so is Y.*
(3) *Suppose Y is locally closed in X. If X is sober then so is Y.*

Proof. Proof of (1). Suppose X is Kolmogorov. Let $x, y \in Y$ with $x \neq y$. Then $\overline{\{x\}} \cap Y = \overline{\{x\}} \neq \overline{\{y\}} = \overline{\{y\}} \cap Y$. Hence $\overline{\{x\}} \cap Y \neq \overline{\{y\}} \cap Y$. This shows that Y is Kolmogorov.

Proof of (2). Suppose X is quasi-sober. It suffices to consider the cases Y is closed and Y is open. First, suppose Y is closed. Let Z be an irreducible closed subset of Y. Then Z is an irreducible closed subset of X. Hence there exists $x \in Z$ with $\overline{\{x\}} = Z$. It follows $\overline{\{x\}} \cap Y = Z$. This shows Y is quasi-sober. Second, suppose Y is open. Let Z be an irreducible closed subset of Y. Then $\overline{Z}$ is an irreducible closed subset of X. Hence there exists $x \in \overline{Z}$ with $\overline{\{x\}} = \overline{Z}$. If $x \notin Y$ we get the contradiction $Z = Z \cap Y \subset \overline{Z} \cap Y = \overline{\{x\}} \cap Y = \emptyset$. Therefore $x \in Y$. It follows $Z = \overline{Z} \cap Y = \overline{\{x\}} \cap Y$. This shows Y is quasi-sober.

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Proof of (3). Immediately from (1) and (2). □